The Last Two Camps

Time limit per test: 2 seconds
Memory limit per test: 1 gb

A traveler is crossing a dangerous frontier consisting of **N** checkpoints arranged in a straight line from west to east.

Each checkpoint has an associated resource value a_i.

Before continuing to the capital, the traveler must establish exactly two camps at two different checkpoints.

If the camps are established at checkpoints **i** and **j** (1 <= **i** < **j** <= **N**), then the efficiency of this arrangement is defined as:

$$(a_j - a_i) + (\mathbf{j} - \mathbf{i})$$

Here, $a_j - a_i$ represents the net gain in resources between the first and second camp, and **j - i** represents the distance traveled between them.

The traveler wishes to maximize the total efficiency. Determine the maximum possible efficiency over all valid choices of two camps.

Input:

The first line contains a single integer **t** representing the number of test cases.

Each test case consists of:

1) A single integer **n** ($2 <= \mathbf{n} <= 2 \times 10^5$) representing the number of checkpoints.

2) A line containing **n** integers $a_1, a_2, ..., a_n$ ($1 <= a_i <= 10^{18}$) representing the resource values of the checkpoints.

It is guaranteed that the sum of n over all test cases does not exceed 2×10^5.

Output:

For each test case, print a single integer: the maximum efficiency achievable.

Example:

Input
3 5 4 2 7 1 9 4 5 5 5 5 3 10 1 20
Output
10 3 20

Explanation:

Test case 1

Choose camps at checkpoints 2 and 5:

$$(a_5 - a_2) + (5 - 2) = (9 - 2) + 3 = 10$$

No other valid pair yields a higher efficiency.

Test case 2

All checkpoints have equal resource value. Thus the score only depends on distance, so choosing checkpoints 1 and 4 gives:

$$(5 - 5) + (4 - 1) = 3$$

Test case 3

Choose camps at checkpoints 2 and 3:

$$(a_3 - a_2) + (3 - 2) = (20 - 1) + 1 = 20$$

Notes:

Camps must be placed at different checkpoints.

The first camp must appear before the second one.